Aufgabe 1: Gleitkommazahlen

Die Formel für Gleitkommazahlen wurde in der Vorlesung folgendermaßen angegeben:

$$\mathrm{f} = (-1)^{\mathrm{s}} \cdot (1 + \mathrm{m}/2^{23}) \cdot 2^{\mathrm{e}-127}$$

Dabei codiert das erste Bit s, die folgenden acht Bits e und die restlichen 23 Bits m. Berechnen Sie, welche Zahlen durch die folgenden Bitmuster dargestellt werden:

a) $1|01111101|10000000000000000000000$ Simulator ↗

$$\begin{aligned}
&= (-1)^{1} \cdot (1 + 10000000000000000000000/2^{23}) \cdot 2^{01111101-127} \\
&= -(1 + 2^{22}/2^{23}) \cdot 2^{125-127} \\
&= -1,5 \cdot 0,25 \\
&= -0,375
\end{aligned}$$

b) $0|10000011|00100000000000000000000$ Simulator ↗

$$\begin{aligned}
&= (-1)^{0} \cdot (1 + 00100000000000000000000/2^{23}) \cdot 2^{10000011-127} \\
&= (1 + 2^{20}/2^{23}) \cdot 2^{131-127} \\
&= 1,125 \cdot 16 \\
&= 18
\end{aligned}$$

c) $1|10000010|10010000000000000000000$ Simulator ↗

$$\begin{aligned}
&= (-1)^{1} \cdot (1 + 10010000000000000000000/2^{23}) \cdot 2^{10000010-127} \\
&= -(1 + 4718592/2^{23}) \cdot 2^{130-127} \\
&= -1,5625 \cdot 8 \\
&= -12,5
\end{aligned}$$

d) $0|10000001|10010000000000000000000$ Simulator ↗

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 10010000000000000000000/2^{23}) \cdot 2^{10000001-127} \\
&= (1 + 4718592/2^{23}) \cdot 2^{129-127} \\
&= 1,5625 \cdot 4 \\
&= 6,25
\end{aligned}
$$

e) $0|01111110|10000000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 10000000000000000000000/2^{23}) \cdot 2^{01111110-127} \\
&= (1 + 2^{22}/2^{23}) \cdot 2^{126-127} \\
&= 1,5 \cdot 0,5 \\
&= 0,75
\end{aligned}
$$

f) $1|10000100|01010100000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 01010100000000000000000/2^{23}) \cdot 2^{10000100-127} \\
&= -(1 + 2752512/2^{23}) \cdot 2^{132-127} \\
&= -1,328125 \cdot 32 \\
&= -42,5
\end{aligned}
$$

g) $0|01111100|10000000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 10000000000000000000000/2^{23}) \cdot 2^{01111100-127} \\
&= (1 + 2^{22}/2^{23}) \cdot 2^{124-127} \\
&= 1,5 \cdot 0,125 \\
&= 0,1875
\end{aligned}
$$

h) $1|01111111|10000000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 10000000000000000000000/2^{23}) \cdot 2^{01111111-127} \\
&= -(1 + 2^{22}/2^{23}) \cdot 2^{127-127} \\
&= -1,5 \cdot 1 \\
&= -1,5
\end{aligned}
$$

i) $0|10000000|01010000000000000000000$ Simulator ↗

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 01010000000000000000000/2^{23}) \cdot 2^{10000000-127} \\
&= (1 + 2621440/2^{23}) \cdot 2^{128-127} \\
&= 1,3125 \cdot 2 \\
&= 2,625
\end{aligned}
$$

j) $1|10000010|00001000000000000000000$ Simulator ↗

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 00001000000000000000000/2^{23}) \cdot 2^{10000010-127} \\
&= -(1 + 2^{18}/2^{23}) \cdot 2^{130-127} \\
&= -1,03125 \cdot 8 \\
&= -8,25
\end{aligned}
$$

k) $1|01111110|01000000000000000000000$ Simulator ↗

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 01000000000000000000000/2^{23}) \cdot 2^{01111110-127} \\
&= -(1 + 2^{21}/2^{23}) \cdot 2^{126-127} \\
&= -1,25 \cdot 0,5 \\
&= -0,625
\end{aligned}
$$

l) $0|10000010|11010000000000000000000$ Simulator ↗

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 11010000000000000000000/2^{23}) \cdot 2^{10000010-127} \\
&= (1 + 6815744/2^{23}) \cdot 2^{130-127} \\
&= 1,8125 \cdot 8 \\
&= 14,5
\end{aligned}
$$

m) $1|10000010|00111000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 00111000000000000000000/2^{23}) \cdot 2^{10000010-127} \\
&= -(1 + 1835008/2^{23}) \cdot 2^{130-127} \\
&= -1,21875 \cdot 8 \\
&= -9,75
\end{aligned}
$$

n) $0|10000000|10010000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 10010000000000000000000/2^{23}) \cdot 2^{10000000-127} \\
&= (1 + 4718592/2^{23}) \cdot 2^{128-127} \\
&= 1,5625 \cdot 2 \\
&= 3,125
\end{aligned}
$$

o) $0|01111110|11000000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 11000000000000000000000/2^{23}) \cdot 2^{01111110-127} \\
&= (1 + 6291456/2^{23}) \cdot 2^{126-127} \\
&= 1,75 \cdot 0,5 \\
&= 0,875
\end{aligned}
$$

p) $1|10000011|00110000000000000000000$ **Simulator ↗**

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 10011000000000000000000/2^{23}) \cdot 2^{10000011-127} \\
&= -(1 + 4980736/2^{23}) \cdot 2^{131-127} \\
&= -1,59375 \cdot 16 \\
&= -25,5
\end{aligned}
$$

q) $0|01111101|01000000000000000000000$ **Simulator** ↗

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 01000000000000000000000/2^{23}) \cdot 2^{01111101-127} \\
&= (1 + 2^{21}/2^{23}) \cdot 2^{125-127} \\
&= 1,25 \cdot 0,25 \\
&= 0,3125
\end{aligned}
$$

r) $1|10000000|10100000000000000000000$ **Simulator** ↗

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 10100000000000000000000/2^{23}) \cdot 2^{10000000-127} \\
&= -(1 + 5242880/2^{23}) \cdot 2^{128-127} \\
&= -1,625 \cdot 2 \\
&= -3,25
\end{aligned}
$$

s) $0|10000001|01100000000000000000000$ **Simulator** ↗

$$
\begin{aligned}
&= (-1)^0 \cdot (1 + 01100000000000000000000/2^{23}) \cdot 2^{10000001-127} \\
&= (1 + 3145728/2^{23}) \cdot 2^{129-127} \\
&= 1,375 \cdot 4 \\
&= 5,5
\end{aligned}
$$

t) $1|10000010|01100000000000000000000$ **Simulator** ↗

$$
\begin{aligned}
&= (-1)^1 \cdot (1 + 01100000000000000000000/2^{23}) \cdot 2^{10000010-127} \\
&= -(1 + 3145728/2^{23}) \cdot 2^{130-127} \\
&= -1,375 \cdot 8 \\
&= -11
\end{aligned}
$$